SPOT EV

PROJECT REPORT

Submitted to

UNIVERSITY OF CALICUT

In partial fulfillment of the requirements of the award for the degree of

BACHELOR OF COMPUTER SCIENCE (2021-2024)

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CERTIFICATE

This is to certify that the project work entitled

SPOT EV

Is the bonafide record of work done by

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Submitted for the viva-voce examination held on……

INTERNAL EXAMINER

EXTERNAL EXAMINER

1.
2.

DECLARATION

We hereby declare that the project report entitled “SPOT EV” done as the partial fulfillment of the requirements for the award of the degree of Bachelor of Science (B.sc COMPUTER SCIENCE), is a project report done by us during the duration of our period of study in Sixth Semester in Le-Ment College of Advanced Studies, Pattambi under the guidance of Mrs.SHAHLA and this has not been previously submitted for the award of Degree in this and other colleges or Universities.

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Place:

Date:

ACKNOWLEDGEMENT

I am grateful to each and every one who has contributed in numerous ways in the development of this project whose efforts have been a healing hand in every footstep towards my destiny.

I give all the glory and honor to the God Almighty, whose divine touch had been the foremost inspiration and guiding light in the fulfillment of this uphill task.

I am glad to present the project report entitled **"SPOT EV"** prepared as a part of Our Sixth Semester, Bachelor of Computer Science program.

I take this opportunity to express my deep sense of gratitude and sincere thanks to our Principal **Mr.SHABEER K.P** of Le-Ment College of Advanced studies for giving us an opportunity to take up the project work.

I place on record our Internal Guide **Mrs.SHAHLA**, faculty of the Computer Science department for her guidance for smooth completion of this project. Without her determination and enthusiasm this project would be incomplete.

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I also express my heartfelt thanks to my dear friends who had supported me in my endeavour to complete this project successfully.

ABSTRACT

The mobile application aims to provide a seamless and convenient experience for users of electric vehicles (EVs) to find and use charging stations. The app utilizes Flutter, a popular mobile development framework, to create an intuitive and user-friendly interface.

Through the app, user can search for nearby charging stations, view information about each station, and receive real-time updates on station availability and charging status. The app also includes features such as the ability to set reminders for when a charging session is completed and the option to pay for charging sessions directly within the app.

The app designed to be highly scalable and adaptable, with the potential to integrate with a variety of different charging station networks and systems. With its sleek and modern design, the app aims to encourage more people to make the switch to electric vehicles by making the charging process as simple and straightforward as possible.

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1. INTRODUCTION

Introducing our Android app for booking charging slots. In today's busy world, finding a Convenient charging slot can often seem difficult. However, with our innovative Android app We aim to simplify the process and revolutionize the way you reserve charging Slots. Our app offers users a hassle-free solution to pre-book charging slots. Whether you're Planning a trip, or an outing, our app enables you to find charging stations and book slots With just a few taps on your smartphone. With Intuitive features and user-friendly interface.

Ensuring a seamless experience from start to finish. Whether you are a frequent traveller or Traveller exploring new destinations, our Android app is your ultimate companion for finding EV charging stations. Say goodbye to endlessly circling the block looking for a spot— Download our app today and experience the convenience of pre-booking your charging Slot. Welcome to the best way to charge EV!

1.1 Objectives:

- Users could input their interested destination, and the app could suggest the efficient route with charging stations along the way.
- The app would maintain a history of all the charging session a user has completed
- User could leave reviews and ratings for charging stations.
- The app allows users to find nearby charging stations and provides information on the type of charging station, availability, and pricing.
- The app provides real-time updates on the station availability, charging status, and estimated charging time.
- The app allows users to pay for charging sessions directly from their mobile device.
- The app allows users to manage their charging account, view charging history, and access customer support

SYSTEM ANALYSIS

2.SYSTEM ANALYSIS

2.1. Existing system

Currently, electric vehicle (EV) owners rely on fragmented solutions to locate charging stations, with standalone apps or websites offering limited integration and real-time updates. The absence of a unified platform means users navigate various interfaces to find charging stations, check availability, and manage payments. This lack of cohesion may result in a less-than-optimal user experience, hindering the widespread adoption of electric vehicles.

2.2Proposed System:

The proposed EV charging system app aims to revolutionize the current landscape by providing a comprehensive and integrated solution. Utilizing Flutter for a seamless user interface, the app will offer a one-stop solution for EV owners, allowing them to easily locate nearby charging stations, receive real-time updates on availability and charging status, and make payments within the app. The design focuses on simplicity and modernity to encourage user adoption. Features such as route planning, reservation systems, and a rewards program are introduced to enhance user convenience and incentivize regular app usage. The app also incorporates social elements to connect EV owners, fostering a sense of community. Overall, the proposed system addresses the limitations of the existing infrastructure, providing a streamlined and user-friendly experience to promote the broader use of electric vehicles

SOFTWARE CONFIGURATION

3. SOFTWARE CONFIGURATION

3.1 SOFTWARE REQUIRMENTS

- Operating System : Android
- Front End : Html, CSS , JavaScript (Web)
  Flutter / Dart (Mobile)
- Back End : PHP
- Connection : TCP / IP
- Protocol : HTTP, SMTP.

3.2 HARDWARE REQUIRMENTS

- Processor : Intel core i3 or above
- Clock Speed : 500 MHZ
- System Bus : 32Bit, 64 Bit
- RAM : Minimum 2GB (recommended 8GB)
- HDD : 20GB
- Monitor : SVGA COLOR
- Keyboard : 108 keys
- Mouse : Any type of mouse

LITERATURESURVEY

4. LITERATURESURVEY

4.1 Feasibility Study

A feasibility study is an evaluation of a proposal designed to determine the difficulty in carrying out a designed task. Generally, a feasibility study precedes technical development and project implementation. In other words, a feasibility study is an evaluation or analysis of the potential impact of a proposed project. Feasibility Study is performed to choose the system that meets the performance requirements at least cost. The most essential tasks performed by a Feasibility Study are the identification and description of candidate systems, the evaluation to determine if developing app is visible and worthwhile. The best system means the system that meet performance requirements at the least cost. The most difficult part of a Feasibility Study is the identification of the candidate systems and the evaluation of their performances and costs. The new system has no additional expense to implement the system. The new system has advantages such as we can easily access files from any client in the Network, accurate output for accurate input and this application is more user friendly. We can use this application not only in this organization but also in other firms. So it is worth solving the problem.

4.2 Technical Feasibility

The technical feasibility of an ecommerce app involves evaluating factors like technology stack, platform compatibility, scalability, security, and integration capabilities. It assesses whether the required technology infrastructure can be implemented effectively to support the app's functionality and performance requirements within budget and time constraints. Additionally, considerations for backend infrastructure, database management, payment gateway integration, and user experience optimization are crucial in determining the technical feasibility of the project.

4.3 Economical Feasibility

Assessing the economic feasibility of an ecommerce app involves considering factors like development costs, potential revenue streams (such as sales commissions, subscription fees, or advertising), market demand, competition, and ongoing operational expenses like maintenance and marketing. Conducting a thorough cost- benefit analysis can help determine whether the projected returns justify the initial investment. Additionally, factors like scalability and long-term sustainability should be taken into account.

4.4 Operational Feasibility

Operational feasibility assesses whether a proposed project or system can be effectively implemented within an organization's existing operations, considering technical requirements, organizational capabilities, impact on day-to-day activities, user acceptance, legal compliance, and overall cost-effectiveness.

4.5 Legal Feasibility

Determines whether the proposed system conflicts with legal requirements or not. e.g.: A data processing system must comply with the local Data Protection Acts.

4.6 Schedule Feasibility

A project will fail if it takes too long to be completed before it is useful. Typically this means estimating how long the system will take to develop, and if it can be completed in a given time period using some methods like payback period. Schedule feasibility is a measure of how reasonable the project timetable is.

4.7 Resource Feasibility

This involves questions such as how much time is available to build the new system, when it can be built, whether it interferes with normal business operations, type and amount of resources required, dependencies, etc. Contingency and mitigation plans should also be stated here.

SYSTEM DESIGN

5. SYSTEM DESIGN

5.1 System architecture

The most creative and challenging phase of the life cycle is system design. The term design describes a final system and the process by which it is developed. It refers to the technical specifications that will be applied in implementations of the candidate system. The design maybe defined as the process of applying various techniques and principles for the purpose of defining a device, a process or a system with sufficient details to permit its physical realization. The designer‘s goal is how the output is to be produced and in what format. Samples of the output and input a real so presented .Second input data and database files have to be designed to meet the requirements of the proposed output. The processing phases are handled through the program Construction and Testing. Finally, details related to justification of the system and an estimate of the impact of the candidate system on the user and the organization are documented and evaluated by management as a step toward implementation. The importance of software design can best a ted in a single word—Quality. Design provides us with representations of software that can be assessed for quality .Design is the only way where we can accurately Translate a customer‘s requirements into a complete software product or system. Without design we risk building an unstable system that might fail if small changes are made. It may as well be difficult to test, or could be one who‘s quality can‘t be tested. So it is an essential phase in the development of a software product.

5.2 Notations used in data flow diagram

In Data Flow Diagrams (DFDs) for an E-Commerce app, various symbols and notations are used to represent different elements and relationships within the system. Here are some common notations used in DFDs for E-Commerce applications:

1. **Process Round (Rectangle):**
   - Represents a function or activity in the system.
   - In an e-Commerce app, processes might include "Order Processing," "Inventory Management," or "Payment Processing.
2. **External Entity (Square):**
   - Represents external entities interacting with the system.
   - In an e-Commerce app, external entities can be "Customer," "Supplier," or "Payment Gateway.
3. **Data store (open-ended Rectangle):**
   - Represents data storage, such as a database or file.
   - In an e-Commerce app, data stores can include "Product Database" or "Customer Information Database.
4. **Data Flow(Arrow):**
   - Represents the flow of data between processes, external entities, and data stores.
   - Arrows indicate the direction of data movement.
   - In an e-Commerce app, data flows can represent the flow of information like "Order Details," "Product Information," or "Payment Data."
5. **Data Flow Source/Destination(oval):**
   - Indicates the origin or destination of data flows.
   - In an e-Commerce app, this could be a process, external entity, or data store.
6. **Control Flow(Dashed line with Arrowhead):**
   - Represents the flow of control or the sequence in which processes occur.

7. **Annotations/Comments:**
    - Used to provide additional information or descriptions for clarity.

This simple DFD represents the flow of data between external entities, processes (such as Order Processing and Inventory Management), and data stores (like a Product Database) in an e-Commerce application. The actual diagram may vary based on the complexity and specific requirements of the system.

5.3 Table Design

1.login_table

Field	Data type	Constraints	Description
Login id	INT	Primary key, Auto-increment	Unique identifier for Each user
Email	VARCHAR(20)	Not null, Unique string	User's Email address.
Password	VARCHAR(20)	Not null, String	User's password

2.user Register_table

Field	Data type	Constraints	Description
Reg_id	INT	Primary key, Auto-increment	Unique identifier for Each user
Name	VARCHAR(20)	Not null, String	User's name
Mobile_no	VARCHAR(20)	Not null, String	User's mobile number
Log_id	INT	Foreign key	User's login id

3.Station_table

Field	Data type	Constraints	Description
Station_ID	INT	Primary key, Auto-increment	Unique identifier for Each charging stations
Name	VARCHAR(20)	Not null, String	Name of the charging station
Location	VARCHAR(20)	Not null, String	Location of the Charging station
Charger Type	VARCHAR(20)	Not null, String	Type of charger at the Station
Availability	BOOLEAN	Boolean	Wheather or not the Station is available
Pricing	DECIMAL	Decimal	Price of charging at the station
Latitude	FLOAT	Float	Latitude of the station
Longitude	FLOAT	Float	Longitude of the station

4.Current Rate_TB

Field	Data type	Constraints	Description
rate_ID	INT	Primary Key, Auto-increment	Unique identifier for each rate (Primary key0
Charger_Type	VARCHAR(50)	Not Null	Type of charger that the rate applies to
rate	DECIMAL(10,2)	Not Null, Greater than Or equal to 0	Current rate for using The charging station
tax	DECIMAL(5,2)	Not Null, Check (tax_percentage>=0)	Tax percentage applied to the rate
Date	DATE	Not Null	Date of the validity period

5.Booking_TB

Field	Data type	Constraints	Description
booking_id	INT	Primary Key	Unique identifier for Each booking
user_id	INT	Primary Key	Identifier of the user who made the booking
station_id	INT	Foreign Key	Identifier of the charging station booked
start_time	DATATIME	Not Null	Start date and time of the booking
end_time	DATATIME	Not Null	End date and time of the booking
Crteated_at	DATATIME	Not Null	Date and time when the booking was created
status	VARCHAR(20)	Not Null	Status of the booking (eg.”pending “,”Confirmed”,”Cancelled”)
payment	INT	Foreign Key	Identifier of the associated payment (if applicable)

6.Complaint_TB

Field	Data type	Constraints	Description
complaint_id	INT	Primary Key	Unique identifier for each complaints
user_id	INT	Foreign Key	Identifier of the user who made the complaint
station_id	INT	Foreign Key	Identifier of the charging station associated with the complaint
timestamp	DATETIME	Not Null	Date and time of the complaint
description	TEXT	Not Null	Description of the complaint
status	VARCHAR(20)	NOT Null	Status of the complaint (eg.”Open”,”Resolved”)

7.Review_TB

Field	Data type	Constraints	Description
review_id	INT	Primary Key	Unique identifier for each review
user_id	INT	Foreign Key	Identifier of the user who wrote the review
station_id	INT	Foreign Key	Identifier of the charging station associated with the review
rating	INT	Not Null, Check(rating >=1 AND rating <=5)	Rating given by the user(1-5 scale)
comment	TEXT	None	Comment or feedback provided by the user
timestamp	DATATIME	Not Null	Date and time when the review was submitted

8.Payment

Field	Data type	Constraints	Description
payment_id	INT	Primary Key	Unique identifier for each payments
user_id	INT	Foreign Key	Identifier of the user who made the payment
ChargingDuration	INTERVAL	Not Null	Duration of the charging session
amount	DECIMAL(10,2)	Not Null, Checked(amount >=0)	The payment amount
payment_date	DATETIME	Not Null	Date and time of the payment
payment_method	VARCHAR(50)	Not Null	The method used for payment, such as credit card, paypal, etc

9.Vehicle_TB

Field	Data type	Constraints	Description
vehicle_id	INT	Primary Key	Unique identifier for each vehicle
user_id	INT	Foreign Key	Identifier of the user who owns the vehicle
vehicle_brand	VARCHAR(50)	Not Null	The brand or manufacture of the vehicle
model	VARCHAR(50)	Not Null	The model name or number of the vehicle
Battery_capacity	INTEGER	Not Null, Check (battery_capacity>0)	The battery capacity of the vehicle in kilowatt-hours(kWh)
vin	VARCHAR(17)	Not Null, Unique	The vehicle identification number (VN) of the vehicle

10.Slot_TB

Field	Data type	Constraints	Description
slot_id	INT	Primary Key	Unique identifier for each slot
station_id	INT	Foreign Key	Identifier of the charging station to which the slot belongs
slot_number	INT	Not Null, Unique	Slot number within the charging station
is_available	BOOLEAN	Not Null	Indicates whether the slot is currently available for charging
connector_type	VARCHAR(50)	Not Null	Type of connector supported by the slot
power_capacity	INT	Not Null,Check (power_capacity>0)	Power capacity of the slot in kilowatt (kWh)

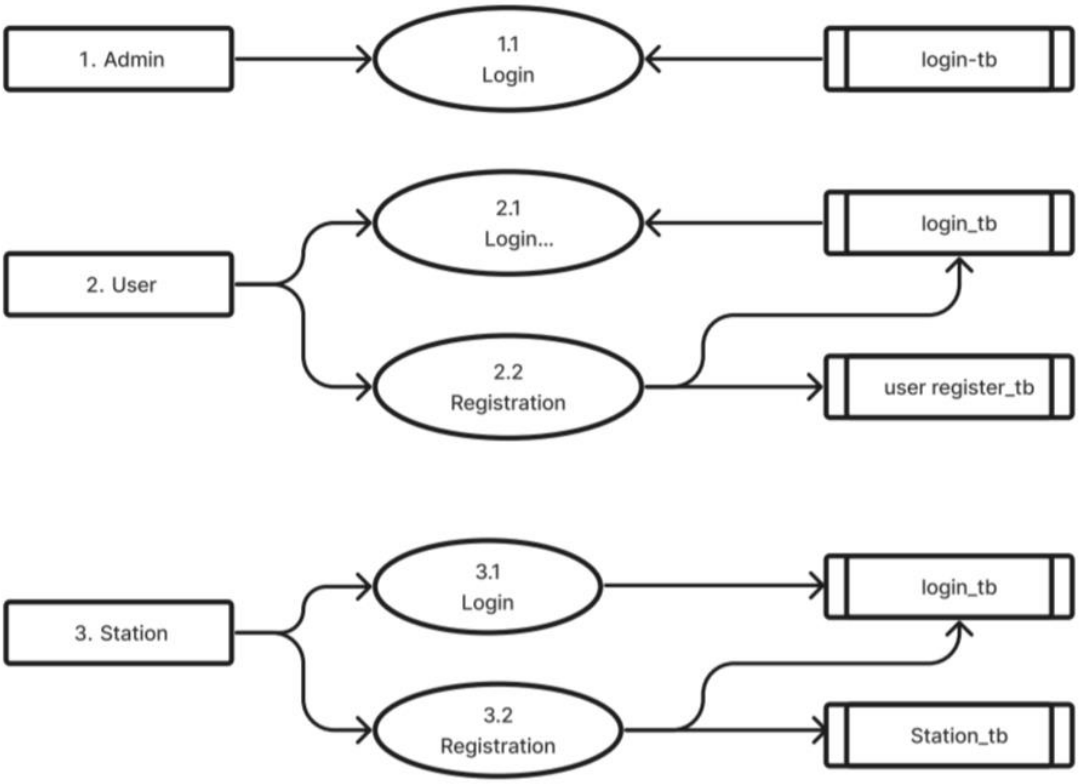
11.Charging_TB

Field	Data type	Constraints	Description
charging_id	INT	Primary Key	Unique identifier for each charging session
user_id	INT	Foreign Key	Identifier of the user who initiated the charging session
station_id	INT	Foreign Key	Identifier of the charging station used for the session
start_time	DATATIME	Not Null	Timestamp of when the charging session started
end_time	DATETIME	None	Timestamp of when the charging session ended
energy_used	DECIMAL(10,2)	Not Null, Checked (energy_used >=0)	The energy consumed during the charging session in kilowatt-hours(kWh)
payment_status	VARCHAR(20)	Not Null, Check (payment_status IN (‘Pending’,’Completed’))	Indicates the payment status of the charging session

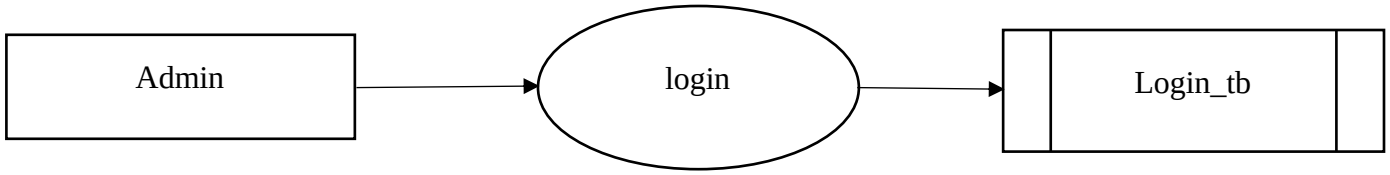
5.4 Constructing a DFD

LEVEL 0

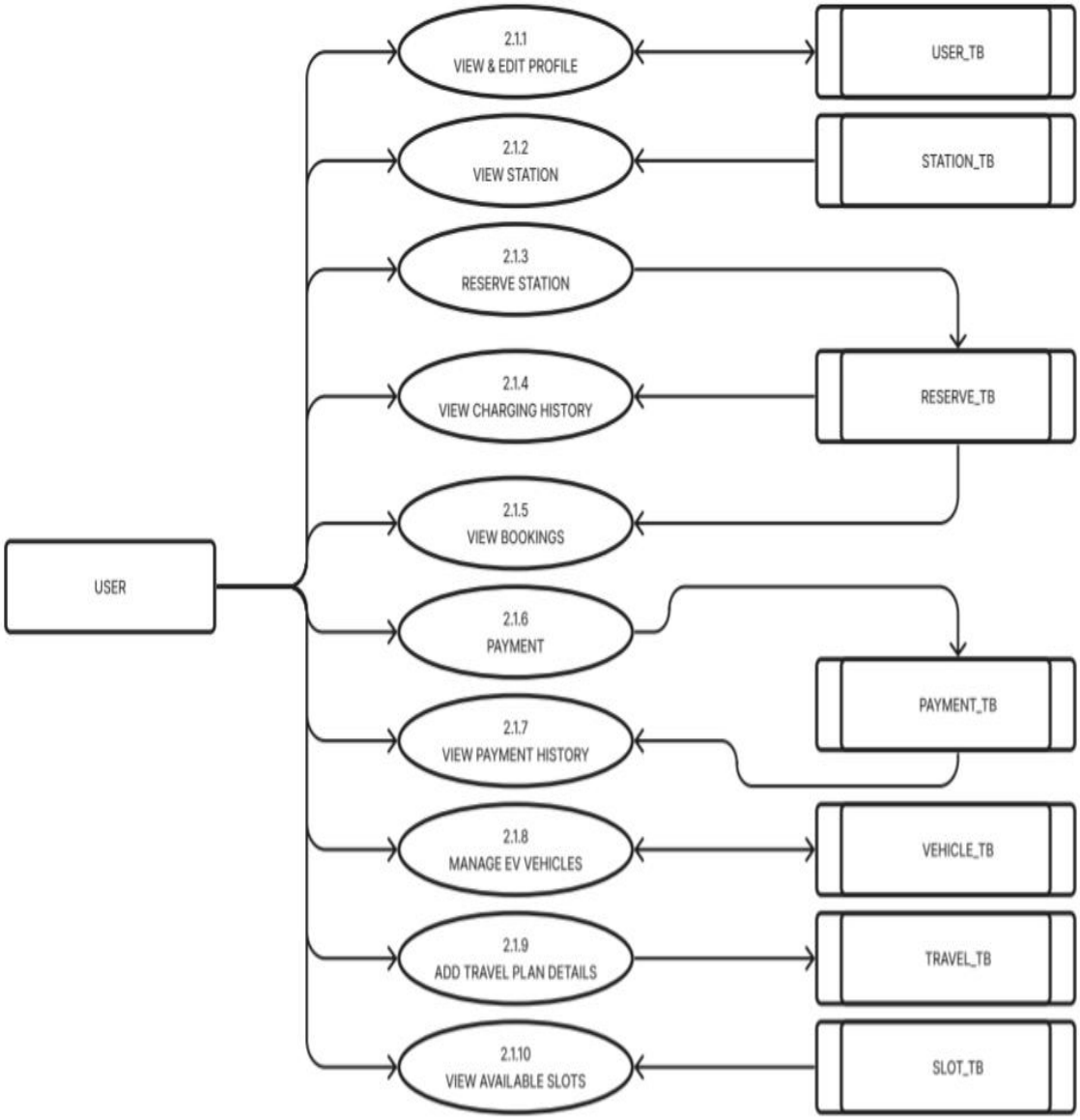
LEVEL 1



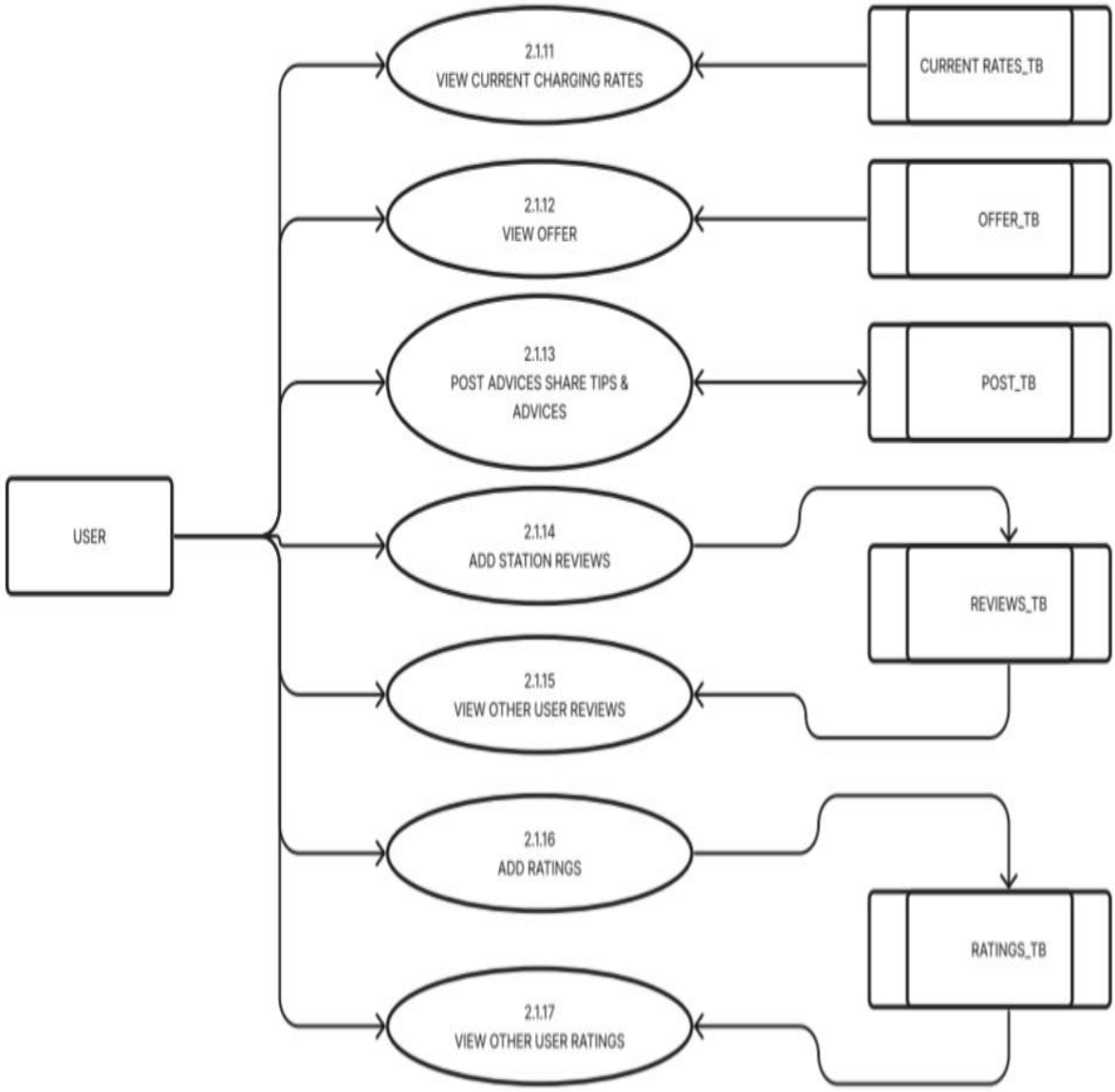
Dataflow diagram for Level 0 and Level 1



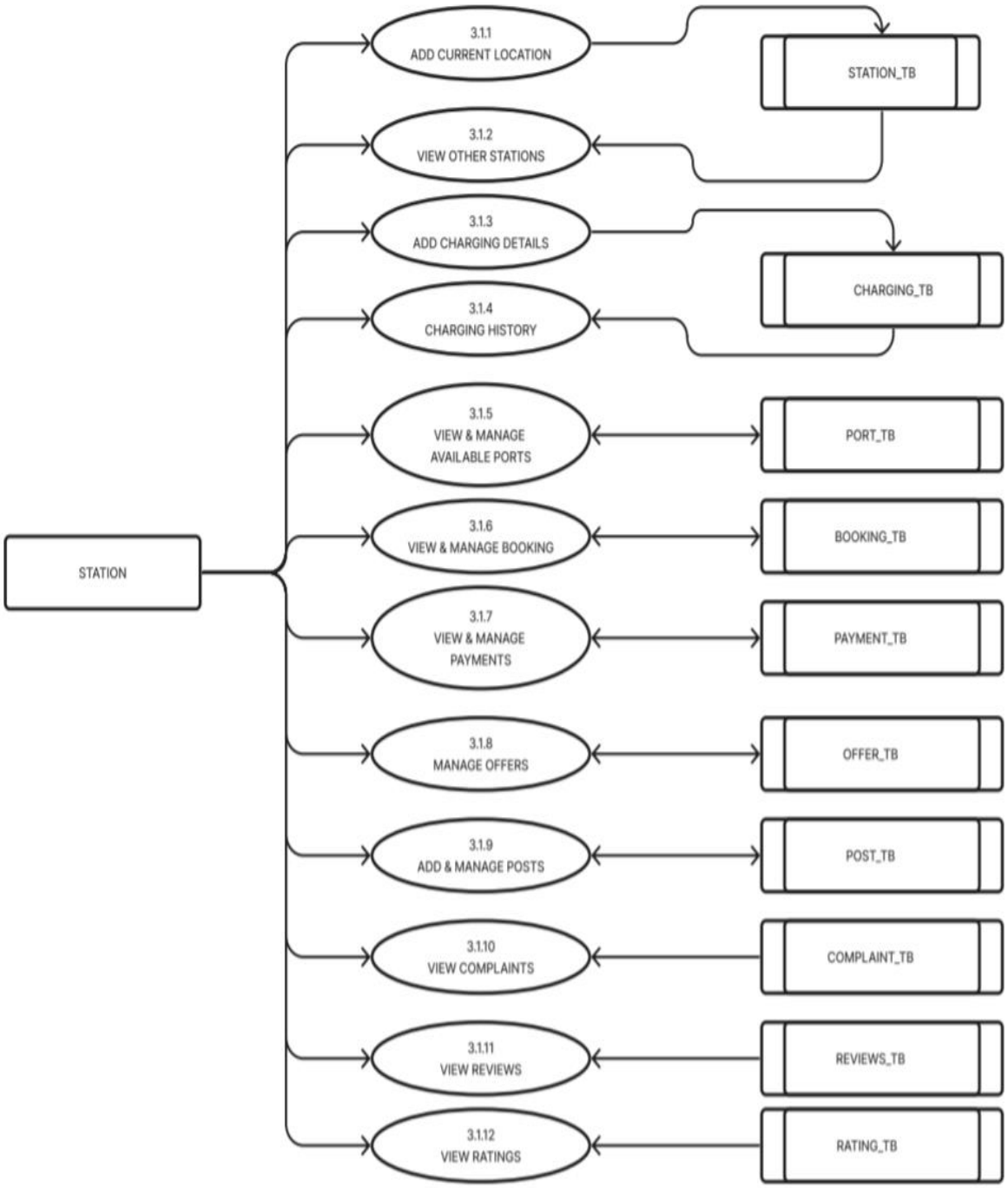
Data Flow Diagram For Admin



Data Flow Diagram For User

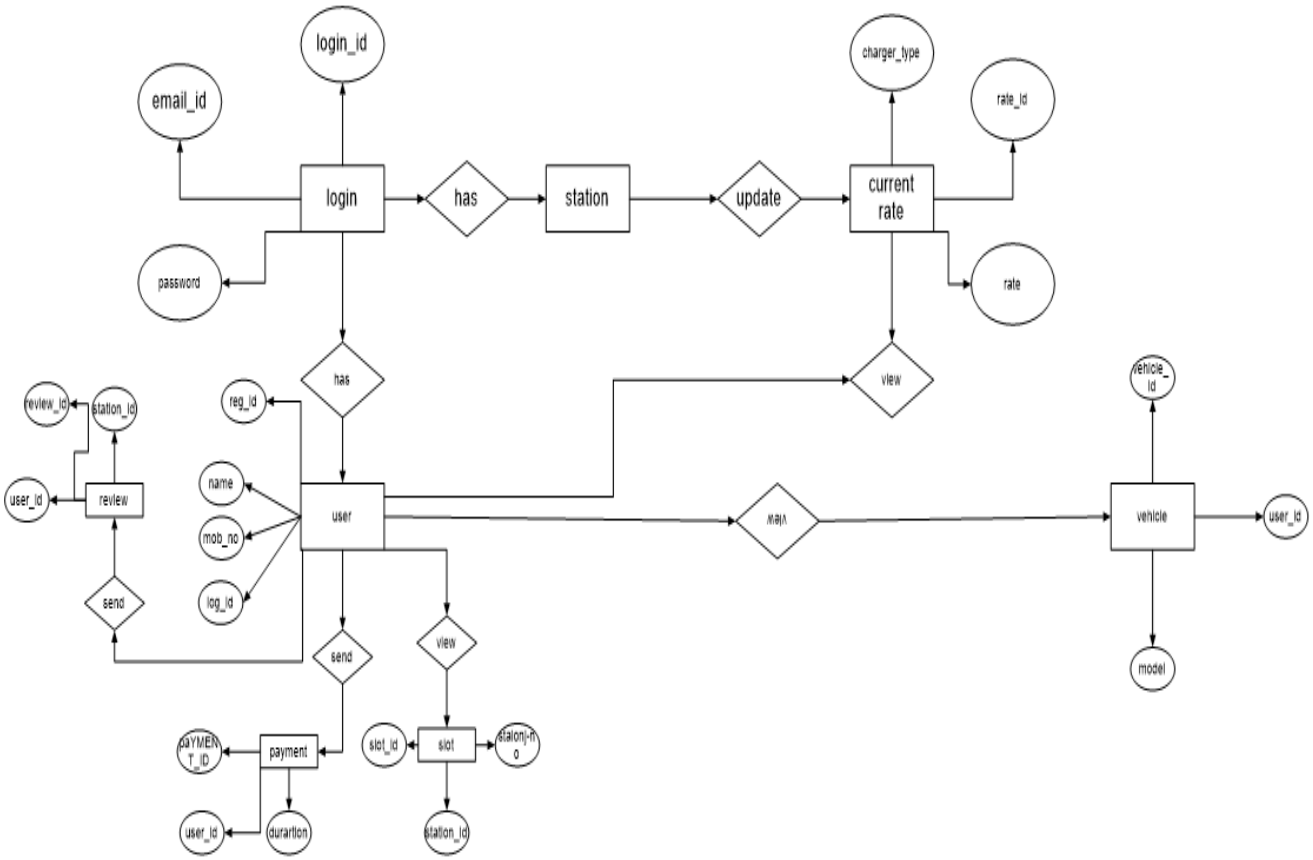


Data Flow Diagram For User



Data Flow Diagram For Station

5.5 ER Diagram



IMPLEMENTATION, TESTING, AND MAINTENANCE

6. IMPLEMENTATION, TESTING, AND MAINTENANCE

6.1 List of modules

- User module
- Admin module
- Station module

User Module:

- Authentication: Allows users to create accounts, log in, and securely manage their personal information.
- Station Search: Enables users to search for nearby charging stations based on their location or intended destination.
- Real-Time Updates: Provides users with real-time information on station availability, charging status, and estimated charging times.
- Route Planning: Allows users to input their destination, and the app suggests the most efficient route with charging stations along the way, considering factors like traffic and weather conditions.
- Reservation System: Enables users to reserve charging stations in advance, ensuring availability upon arrival.

- Charging History: Maintains a history of all charging sessions, including start and end times, costs, and relevant details.
- Payment: Allows users to pay for charging sessions directly within the app.
- Reminders: Users can set reminders for when a charging session is complete.
- Charging Rewards: Provides incentives such as rewards or discounts for using the app to charge their EV.
- Social Features: Allows users to connect with other EV owners, share tips, advice, and arrange meetups.

Station Module:

- Station Locator: Provides information on the location of charging stations, including type, availability, and pricing.
- Real-Time Updates: Updates the station's status in real-time, notifying users of availability and charging progress.
- Reservation Management: Allows station operators to manage reservations made by users.
- Payment Processing: Handles the payment process for charging sessions, ensuring secure transactions.
- Reviews and Ratings: Displays user reviews and ratings for the charging station, helping users make informed decisions.
- Account Management: Allows station operators to manage their station's profile, view usage statistics, and access customer support.

Admin Module:

- User Management : Enables administrators to manage user accounts, handle authentication, and address account-related issues.

- Station Management : Provides tools for administrators to add, update, or remove charging stations, ensuring accurate and up-to-date information.

- Payment Management: Monitors and manages payment transactions, ensuring financial accuracy and security.

- Analytics : Offers insights into app usage, charging patterns, and other relevant analytics for strategic decision-making.

- Customer Support : Provides a platform for administrators to address user or station operator inquiries and concerns.

- System Configuration : Allows administrators to configure system settings, integrate with different charging station networks, and ensure scalability and adaptability.

6.2 Introduction programming languages, IDE's, Tools Technologies

Html

HTML (Hypertext Mark-up Language) is a text-based approach to describing how content contained within an HTML file is structured. This mark-up tells a web browser how to display text, images and other forms of multimedia on a webpage.

HTML is a formal recommendation by the World Wide Web Consortium (W3C) and is generally adhered to by all major web browsers, including both desktop and mobile web browsers. HTML5 is the latest version of the specification.

HTML is a text file containing specific syntax, file and naming conventions that show the computer and the web server that it is in HTML and should be read as such. By applying these HTML conventions to a text file in virtually any text editor, a user can write and design a basic webpage, and then upload it to the internet.

The most basic of HTML conventions is the inclusion of a document type declaration at the beginning of the text file. This always comes first in the document, because it is the piece that affirmatively informs a computer that this is an HTML file. The document header typically looks like this: <!DOCTYPE html>. It should always be written that way, without any content inside it or breaking it up. Any content that comes before this declaration will not be recognized as HTML by a computer.

Doc types are not just used for HTML, they can apply to the creation of any document that uses SGML (Standard Generalized Mark-up Language). SGML is a standard for specifying a specific mark-up language being used. HTML is one of several mark-up languages that SGML and doc type declarations apply to.

The other critical requirement for creating an HTML file is saving it with a .html file extension. Whereas the doc type declaration signals HTML to the computer from the inside of the file, the file extension signals HTML to the computer from the outside of the file. By having both, a computer can tell that it's an HTML file whether it's reading the file or not. This becomes especially important when uploading the files to the web, because the web server needs to know what to do with the files before it can send them to a client computer for the inner contents to be read.

Flutter

Flutter is a UI framework for creating native mobile apps. It allows developers to build a mobile app using a single codebase while helping quickly build iOS and Android apps.

The Flutter framework is a UI toolkit that helps developers create high-performance, high-fidelity applications for iOS, Android, Linux, Mac, Windows, etc. At its core, it lets developers build an application UI using widgets that have different states and react to specific events.

Flutter framework, powered by Dart (programming language), enables maintaining a single codebase for deployment across the platforms mentioned above. This translates to a smooth, responsive application UI across different screen types, aspect ratios, and orientations.

the Flutter framework emulates the powers of core languages like Swift, Java, and JavaScript. This way, every application developed appears native to each platform. This creates a smooth app and reduces the time taken for development and maintenance by a significant factor.

PHP

PHP (Hypertext Processor) is a general-purpose scripting language and interpreter that is freely available and widely used for web development. The language is used primarily for server-side scripting, although it can also be used for command-line scripting and, to a limited degree, desktop applications. The acronym PHP was originally derived from Personal Home Page Tools, but it now

stands for PHP: Hypertext Pre-processor, which the PHP Group's documentation describes as a "recursive acronym."

When used for server-side scripting, PHP is added to a webpage for the purpose of generating dynamic content when the page is accessed through a client browser. The web server runs the script before transmitting the page to the browser. To support this process, the web server requires PHP to be installed on the server, along with a PHP parser -- either a Common Gateway Interface (CGI) parser or a server module.

When a user requests a webpage from the server, the parser interprets the PHP portion of the page, performs the operations called for in the PHP script, and generates the Hypertext Mark-up Language (HTML) that results from those operations. The HTML is then sent to the client browser, along with any other HTML on the page, providing a seamless rendering of the content. Webpages that contain PHP script are considered to be dynamic HTML pages because content varies based on the results of interpreting the script.

CSS

Cascading Style Sheets (CSS) is a style sheet language used for specifying the presentation and styling of a document written in a mark-up language such as HTML or XML (including XML dialects such as SVG, MathML or XHTML).CSS is a cornerstone technology of the World Wide Web, alongside.

CSS is designed to enable the separation of content and presentation, including layout, colours, and fonts.[3] This separation can improve content accessibility;[further explanation needed] provide more flexibility and control in the specification of presentation characteristics; enable multiple web pages to share formatting by specifying the relevant CSS in a separate .CSS file, which reduces complexity and repetition in the structural content; and enable the .CSS file to be cached to improve the page load speed between the pages that share the file and its formatting.

Separation of formatting and content also makes it feasible to present the same mark-up page in different styles for different rendering methods, such as on-screen, in print, by voice (via speech-based browser or screen reader), and on Braille-based tactile devices. CSS also has rules for alternate formatting if the content is accessed on a mobile device.

Java script

JavaScript is a dynamic programming language that's used for web development, in web applications, for game development, and lots more. It allows you to implement dynamic features on web pages that cannot be done with only HTML and CSS.

Many browsers use JavaScript as a scripting language for doing dynamic things on the web. Any time you see a click-to-show dropdown menu, extra content added to a page, and dynamically changing element colours on a page, to name a few features, you're seeing the effects of JavaScript.

6.3 IDE's Tools & Technologies

Android studio

Android Studio is the official integrated development environment (IDE) for Android app development, developed by Google. It provides a comprehensive suite of tools and features to help developers create feature-rich Android applications.

Android Studio is built on top of the Intelligence IDEA community edition and provides a range of features as code completion, syntax highlighting, debugging, testing, and profiling. It also includes an emulator that allows developers to test their applications on a range of virtual devices with different configurations.

One of the features of Android Studio is its support for the Android SDK, which includes libraries and APIs for developing Android applications. Android Studio provides a graphical user interface for. managing the SDK and it into the development process

Android Studio also provides tools for building and signing Android application packages (APKs), which can be distributed through the Google Play Store or other app stores. It includes a range of templates and wizards to help developers get started quickly and provides access to a range of third-party libraries and tools through its integrated Gradle build system

Overall, Android Studio is a powerful and versatile development environment for Android app It provides a range of tools and features to help developers create high-quality, feature-rich Android applications, and is the recommended IDE for Android development by Google

6.4 Test Plan & Test Activities

A plan is necessary for any project to plan the less phase and decide the scope of the project

Test plan involves collecting design specifications about the project, wiring test cases, executing them manually or automatically using automated testing tools. Testing any application is highly important. A test planning same of documenting the test cases, specification plans, and other bene level details about how the application works Test Activities for this works.

Test activities for this project includes various testing like:

- Black Box testing: In this project, sample test cases are written and manual testing is done to check the functionality of the web application.

- White Box testing: Once the web application meets the user requirements and functionalities wording to the test cases, its internal logic is completely tested to ensure that the web application does not have any logical errors or issues.

- Unit testing we have tested all the modules of the web application individually by running a test program.

- Integration testing: After testing the modules individually, tested them by integrating all the sub modules, modules into one web app

- System Testing: It refers to checking whether the system in which the web application is built the necessary requirements like software support.

- End to End Testing: Tested the complete environment of the web application by connecting the device with different machines, installing it as an APK file, with the database, and in the local network

- Usability Testing: Finally, usability testing is performed by testing the web application's flow, UI design, and how flexible and easy the web application is to us.

6.5 Code

```
import'dart:convert';

import 'package:flutter/cupertino.dart';
import 'package:flutter/material.dart';
import 'package:http/http.dart';
import 'package:shared_preferences/shared_preferences.dart';
import 'package:spot_ev/Screens/User/navBar/home/homePage.dart';
import 'package:spot_ev/Screens/connect.dart';


import 'Screens/Station/BottomNavBarPage.dart';
import 'Screens/Station/home/HomePage.dart';
import 'Screens/User/navBar/navBar.dart';
import 'Screens/User/navBar/stations/map.dart';
import 'Screens/signup.dart';
import 'Screens/forgetPassword.dart';
import 'Screens/styles/textstyle.dart';

class LoginPage extends StatefulWidget {
  LoginPage({Key? key, required this.type}) : super(key: key);
  var type;
```

```
  @override
  State<LoginPage> createState() => _LoginPageState();
}



class _LoginPageState extends State<LoginPage> {
  Future<void> saveLoginId(String LoginID)  async {
     SharedPreferences prefs=await SharedPreferences.getInstance();

   prefs.setString('LoginID',LoginID);



  }
  var visible = true;
  var mail = TextEditingController();
  var pass = TextEditingController();
  final formkey = GlobalKey<FormState>();
  Future<void> sendData() async {
    var data = {
      'email': mail.text,
      'password': pass.text,
      'user_type': widget.type,
    };
    var response = await post(Uri.parse('${con.url}/login.php'), body: data);
    print(response.body);
    print(jsonDecode(response.body));
    var LoginID;
    LoginID=jsonDecode(response.body)['log_id'];
```

```
  print('.....$LoginID');
  if(LoginID!=null)
  saveLoginId(LoginID);
  if (jsonDecode(response.body)['result'] == 'User successfully login') {
    final spref = await SharedPreferences.getInstance();
      spref.setString('regi_id', LoginID);
    if (widget.type == 'user') {
      Navigator.pushReplacement(
          context, MaterialPageRoute(builder: (context) =>BottomNavBar()));
          // Nature()
          //  BottomNavBar()));
      ScaffoldMessenger.of(context)
          .showSnackBar(SnackBar(content: Text('user login')));
    }
    if (widget.type == 'station') {
      Navigator.pushReplacement(
          context, MaterialPageRoute(builder: (context) => StationBottomNavBArPage()));
      ScaffoldMessenger.of(context)
          .showSnackBar(SnackBar(content: Text('station login')));
    }
  } else {
    ScaffoldMessenger.of(context)
        .showSnackBar(SnackBar(content: Text('Invalid Credential')));
    Navigator.pushReplacement(
        context,
        MaterialPageRoute(
            builder: (context) => LoginPage(type: widget.type)));
  }
}
```

```
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: SafeArea(
        child: SingleChildScrollView(
          child: Padding(
            padding: const EdgeInsets.all(8.0),
            child: Column(
              mainAxisAlignment: MainAxisAlignment.center,
              children: [
              SizedBox(height: 50,),
              Container(
                height: 100,
                width: 170,
                decoration: BoxDecoration(
                    image: DecorationImage(
                        image: AssetImage('assets/logoo.png'),
                        fit: BoxFit.contain)),
              ),
               SizedBox(
                height: 30,
              ),
              Text(
                'Sign In',
                style: TextStyle(color: Color.fromARGB(255, 101, 3, 153),fontWeight:
FontWeight.bold,fontSize: 20),
              ),
              SizedBox(
                height: 30,
```

```
),
Form(
    key: formkey,
    child: Column(
        mainAxisAlignment: MainAxisAlignment.start,
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          Text(
            '    Email',
            style: email,
          ),
          Padding(
            padding: const EdgeInsets.all(8.0),
            child: Card(
              elevation: 10,
              color: Colors.black,
              child: TextFormField(
                keyboardType: TextInputType.emailAddress,
                controller: mail,
                decoration: InputDecoration(
                  enabledBorder: OutlineInputBorder(
                      borderSide: BorderSide.none),
                  filled: true,
                  fillColor: Colors.grey[350],
                  hintText: 'ENTER YOUR EMAIL',
                  prefixIcon: Icon(
                    Icons.mail_outline_rounded,
                  ),
                ),
```

```
    ),
  ),
),
SizedBox(
  height: 10,
),
Text(
  '    Password',
  style: email,
),
Padding(
  padding: const EdgeInsets.all(8.0),
  child: Card(
    elevation: 10,
    color: Colors.black,
    child: TextFormField(
      obscureText: visible,
      controller: pass,
      decoration: InputDecoration(
        enabledBorder: OutlineInputBorder(
            borderSide: BorderSide.none),
        filled: true,
        fillColor: Colors.grey[350],
        suffixIcon: IconButton(
            onPressed: () {
              setState(() {
                visible = !visible;
              });
            },
```

```
            icon: visible
                ? Icon(CupertinoIcons.eye_slash_fill)
                : Icon(CupertinoIcons.eye)),
        hintText: 'ENTER YOUR PASSWORD',
        prefixIcon: Icon(
          Icons.lock_outline_rounded,
        ),
      ),
    ),
  ),
),
// Padding(
//   padding: const EdgeInsets.only(left: 200),
//   child: Center(
//       child: TextButton(
//           onPressed: () {
//             Navigator.push(
//                 context,
//                 MaterialPageRoute(
//                     builder: (context) =>
//                         ForgetPassword()));
//           },
//           child: Text(
//             'Forgot Password ?',
//           ))),
// ),
SizedBox(
  height: 40,
),
```

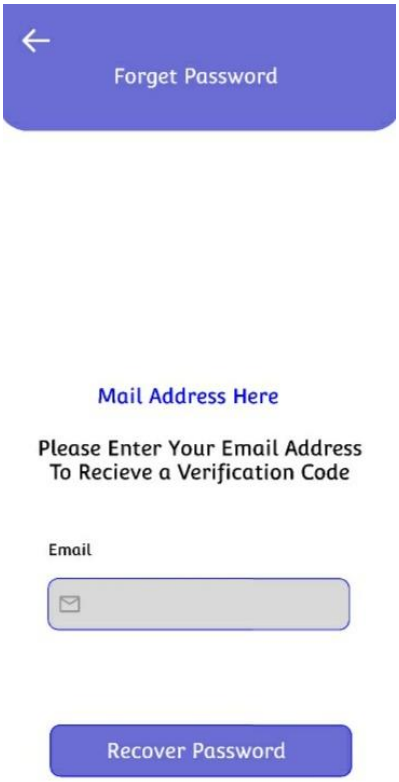
```
Center(
  child: Column(
    children: [
      Container(
        height: 50,
        width: 320,
        child: ElevatedButton(
            style: ElevatedButton.styleFrom(
              shape: RoundedRectangleBorder(
                  borderRadius:
                      BorderRadius.circular(20)),elevation: 20,
              backgroundColor: Color.fromARGB(255, 101, 3, 153),
            ),
            onPressed: () {
              print(mail.text);
              print(pass.text);
              if (mail.text.isNotEmpty &&
                  pass.text.isNotEmpty) {
                print('inside if');
                setState(() {
                  sendData();
                });
              } else {
                ScaffoldMessenger.of(context)
                    .showSnackBar(SnackBar(
                        content:
                            Text('All fields required')));
                Navigator.pushReplacement(
                    context,
```

```
                    MaterialPageRoute(
                        builder: (context) => LoginPage(
                            type: widget.type)));
                }

                // Navigator.push(context, MaterialPageRoute(builder:
(context)=>BottomNavBar()));
              },
              child: Text(
                'LOGIN',
                style: TextStyle(color: Colors.white),
              )),
            ),

            SizedBox(
              height: 20,
            ),

            Row(
              mainAxisAlignment: MainAxisAlignment.center,
              crossAxisAlignment: CrossAxisAlignment.center,
              children: [
                SizedBox(
                  height: 60,
                ),
                Text(
                  'Don\'t Have an Account?',
                  style: email,
                ),
```

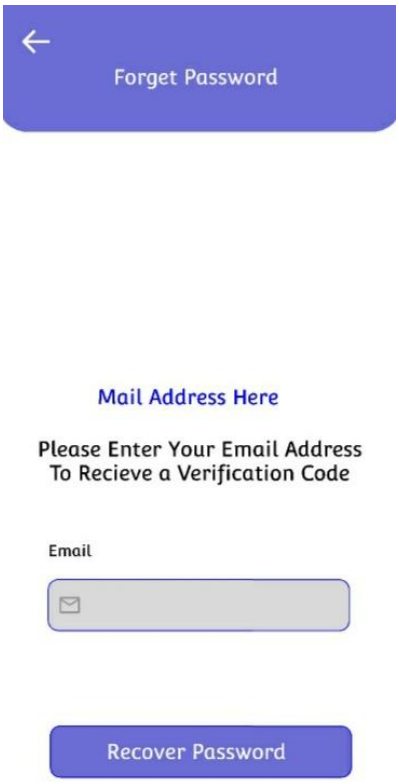
```
                    TextButton(
                      onPressed: () {
                        Navigator.push(
                          context,
                          MaterialPageRoute(
                            builder: (context) => signUPPage(
                              type: widget.type,
                            )));
                      },
                      child: Text(
                        'Sign Up',style: TextStyle(color: Color.fromARGB(255, 101, 3, 153)),
                      ))
                  ],
                ),
              ],
            ),
          )
        ])),
      ]),
    ),
  ),
),
);
}
}
```

6.6 Webpage screenshots

Login page of user



Login page of station



Login page of admin

SCOPE FOR FUTURE WORK

7. SCOPE FOR FUTURE WORK

The ‘SPOT EV’ charging app can be further improved and expanded to offer additional features and functionalities to uses. Here are some potential areas further development.

Reservation System : The app can be expanded to include reservation system. This feature will allow users to reserve charging stations in advanced, ensuring that they are available when they arrive. This could be especially useful for long-distance trips, where charging stations may be high demand.

Charging Rewards : The app can be expanded to include charging rewards. This feature will allow users to earn rewards or discounts for using the app to charge their EV. This could incentivize users to use the app more frequently and help to promote the adoption of EVs.

Social Features : The app can be expanded to include social features such as the ability to connect with other EV owners, share tips and advice, and arrange meet-ups. This could help to foster a sense of community among EV owners and encourage more people to switch to electric vehicles.

Trip Planning : The app can be expanded to include trip planning. This feature will allow users to plan their EV charging stops on a route to ensure they have enough battery charge to reach their destination.

CONCLUSION

8. CONCLUSION

In conclusion, the SPOT EV charger is a web based mobile application that provides user with an easy and convenient way to search for nearby EV charging stations. The app is designed to be user-friendly and intuitive, with a simple and straightforward user interface that allows users to search for charging stations.

The app is designed using Flutter, the popular mobile development framework, and database with a front- end developed using HTML and CSS(web)

The project has demonstrated the feasibility of developing user-friendly and reliable system for searching EV charging stations. The project's outcomes have met the requirements and expectations set out in the project scope, with a fully functional system developed and tested.

There are still potential areas for further development and expansion of the system, including